

1. Determine the processing sequence and the performance measure given below that would result from priority rules **SPT** & **EDD**. Compare the performance measures. (7)

a. Average Completion Time

b. Utilization

c. Average No. of Jobs in the System

d. Average Job Lateness

Job	Job Time (hour)	Due Date (hour)
a	14	20
b	3	8
c	12	15
d	6	24
e	7	6

Answer: For priority rule **SPT**:

Job	Job Time (hour)	Flow Time	Job Due Date (hour)	Job Lateness
b	3	3	8	0
d	6	9	24	0
e	7	16	6	10
c	12	28	15	13
a	14	42	20	22
	Total= 42	Total= 98		Total= 45

1. Average completion time = (Sum of total flow time/No. of jobs) = $98/5 = 19.6$ hour

2. Utilization = (Total job work time/Sum of total flow time) = $42/98 = 42.85\%$

3. Average number of jobs in the system = (Sum of total flow time/Total job work time) = $98/42 = 2.33$

4. Average job lateness = (Total lateness/No of jobs) = $45/5 = 9$ hour

For priority rule **EDD**:

Job	Job Time (hour)	Flow Time	Job Due Date (hour)	Job Lateness
e	7	7	6	1
b	3	10	8	2
c	12	22	15	7
a	14	36	20	16
d	6	42	24	18
	Total= 42	Total= 117		Total= 44

1. Average completion time = (Sum of total flow time/No. of jobs) = $117/5 = 23.4$ hour
2. Utilization = (Total job work time/Sum of total flow time) = $42/117 = 35.89\%$
3. Average number of jobs in the system = (Sum of total flow time/Total job work time) = $117/42 = 2.78$
4. Average job lateness = (Total lateness/No of jobs) = $44/5 = 8.8$ hour

(Your comment on performance measures)

2. A time study was conducted on a task that consists of four elements. The observations and performance ratings for six cycles are shown in the table below. Calculate standard time for this task assuming an allowance factor of 15%. (6)

Element	Performance Rating	Observations (minute per cycle)					
		1	2	3	4	5	6
A	90%	0.44	0.55	0.43	0.45	0.48	0.46
B	85%	1.50	1.54	1.47	1.51	1.49	1.52
C	110%	0.84	0.88	0.77	0.83	0.85	0.87
D	100%	1.10	1.14	1.08	1.12	1.16	1.26

Answer:

Average observation for element A = $(0.44+0.43+0.45+0.48+0.46)/5 = 0.452\text{min}$

Average observation for element B = $(1.47+1.49+1.50+1.51+1.52+1.54)/6 = 1.505\text{min}$

Average observation for element C = $(0.84+0.88+0.83+0.85+0.87)/5 = 0.854\text{min}$

Average observation for element D = $(1.08+1.10+1.12+1.14+1.16)/5 = 1.12\text{min}$

Normal Time for element A = $(0.452 \times 0.9) = 0.406\text{min}$

Normal Time for element B = $(1.505 \times 0.85) = 1.279\text{min}$

Normal Time for element C = $(0.854 \times 1.1) = 0.939\text{min}$

Normal Time for element D = $(1.12 \times 1) = 1.12\text{min}$

Total Normal Time = $(0.406+1.279+0.939+1.12) = 3.745\text{min}$

Standard Time = $3.745 / (1 - 0.15) = 4.406\text{min}$

1. Use Johnson's Rule to obtain the optimum sequence for processing the jobs shown in the figure below. Find out the total completion time and idle time of the machines. (6)

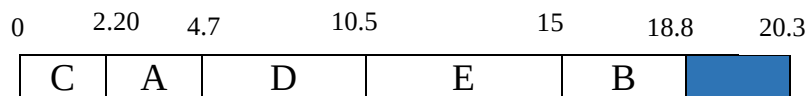
Job times (hours)		
Job	Work Center A	Work Center B
A	2.50	4.20
B	3.80	1.50
C	2.20	3.00
D	5.80	4.00
E	4.50	2.00

Answer:

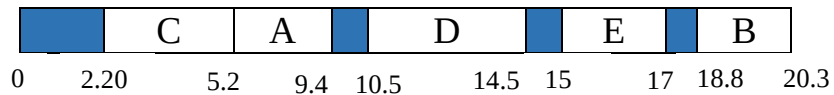
Sequence: $C \rightarrow A \rightarrow D \rightarrow E \rightarrow B$

Gantt Chart :

W/C1:



W/C 2:



Total Completion time: 20.3 hours

Total Idle time: $1.5 + 2.2 + 1.1 + 0.5 + 1.8 = 7.1$ hours